Vaibhav Garg

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I am a B. Tech student from GGSIPU with strong proficiency in Python development, data analytics and data science. I am an enthusiastic software developer who combines technical expertise with an analytical mindset to create effective solutions. I am keen to contribute to innovative projects and grow in fast paced, technology-oriented environments.

EDUCATION

GURU GOBIND SINGH INDRAPRASTHA UNIVERSITY New Delhi

Dr. Akhilesh Das Gupta Institute of Professional Studies Delhi

Bachelor of Technology in Computer Science 2026

Final Year

PROJECTS

Semantic Video Search & Q&A Engine (RAG Pipeline) | *Python, OpenAI Whisper, Scikit-Learn*

- Developed an end-to-end Retrieval Augmented Generation (RAG) system to enable natural language querying of video course content, returning precise timestamps and summaries.
- Built an automated data ingestion pipeline using FFmpeg for audio extraction and OpenAI Whisper (Large-v2) for high-accuracy speech-to-text transcription.
- Engineered a custom semantic search mechanism using BGE-M3 embeddings and Scikit-Learn to calculate Cosine Similarity, manually implementing vector retrieval logic to optimize search precision.
- Integrated Llama 3.1 (via Ollama) to synthesize retrieved context into human-like responses, effectively bridging the gap between unstructured video data and user queries**.**

Jarvis – Voice-Controlled Virtual Assistant (Python)

- Developed using Python, enabling seamless interaction through speech-based commands.
- Integrated libraries such as speech recognition, pyttsx3, and gTTS for real-time audio input processing and voice response generation with <1 second latency.
- Automated 10+ daily tasks, including launching websites, fetching live news via APIs, and holding dynamic conversations powered by GPT.
- Implemented and worked with 4+ APIs and libraries, applying event-driven programming for responsive and real-time task execution.
- Enabled user-friendly voice interaction by combining natural language processing (NLP) and system control logic.
- Demonstrated practical experience in building AI-powered assistants with real-world application potential.

Liver Damage Prediction using Logistic Regression And KNN (Link)

- Integrated Logistic Regression and K-Nearest Neighbors (KNN) algorithms for supervised classification.
- Achieved an accuracy of 86% using the KNN model after hyperparameter tuning and performance evaluation.
- Performed comprehensive data preprocessing, including handling missing values, label encoding, and feature normalization.
- Conducted exploratory data analysis (EDA) to uncover patterns and anomalies in the dataset using Pandas, Seaborn, and Matplotlib.
- Created heatmaps, boxplots, and scatter plots to analyse feature correlations and detect outliers.
- Evaluated model performance using metrics such as accuracy, confusion matrix, precision, recall, and F1-score.
- Used train-test split and cross-validation to ensure the model's robustness and prevent overfitting.

TECHNICAL PROFICIENCIES / SKILLS

CERTIFICATIONS – Verified Credentials

DSA in C++ Coding Blocks (Oct 2023)

Data Science & Machine Learning – IBM (Sept 2024)

Data Analytics – WS Cube Tech (March 2025)

Data Analytics Job Simulation – Deloitte (May 2025)

TECHNOLOGIES

Languages: Python, C++, SQL, JAVA

Tools & Frameworks: Excel, SQL, Pandas, Data Visualization, Power BI, Git version control

AI & Machine Learning: Gen AI, RAG pipelines, Embeddings, LLM, Scikit-learn, Regression & Classification Models

Web Development & Designing: HTML, CSS, JavaScript, UI/UX

Data Structures & Algorithms: Proficient in DSA with C++